

A Midterm Progress Report
On
Real-Time ChatApp

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ABSTRACT

Real-time communication has become an essential part of modern digital interaction. Traditional messaging systems often suffer from delays, lack of instant delivery, and limited features. This project presents the development of a Real-Time Chat Application that enables instant communication between users through a secure and feature-rich platform.

The proposed system is developed using the MERN Stack (MongoDB, Express.js, React.js, Node.js) along with Socket.IO technology for real-time bidirectional communication. The application allows users to register, log in securely using JWT authentication, and exchange text messages, images, and voice messages in real time.

Key features implemented in the system include online/offline user status indicators, typing indicators, message read receipts, message editing, message deletion, complete chat deletion, user search functionality, and unread message counters.

This mid-term report presents the current development status of the project, highlighting the successful completion of the first two objectives: development of a secure real-time chat application and implementation of core messaging features. The third objective, which focuses on secure user authentication, profile management, and password recovery, is currently in progress and will be completed in the final phase of development.

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CHAPTER 1 INTRODUCTION

1.1 Introduction to Project

Real-time communication has become an essential part of modern digital interaction. With the increasing use of online communication platforms, users expect fast, secure, and reliable messaging systems that allow instant interaction without delays. Traditional communication methods such as emails and SMS lack real-time capabilities and often suffer from latency issues, making them unsuitable for instant messaging needs.

The Real-Time Chat Application is a modern web-based communication platform developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The primary purpose of this application is to enable users to communicate instantly through real-time messaging over the internet. The application provides a secure and interactive environment where registered users can exchange text messages, share images, send voice messages, and manage their conversations efficiently.

To achieve real-time communication, the system uses Socket.IO technology, which establishes a persistent bidirectional connection between the client and server. This enables messages to be delivered instantly without requiring page refreshes. Unlike traditional HTTP requests where the client must poll the server for new messages, Socket.IO allows the server to push messages to clients as soon as they are sent, resulting in near-instantaneous delivery.

The application includes advanced features that enhance the user experience. Online and offline status indicators show users which contacts are currently active. Typing indicators notify users when someone is composing a message, creating a more interactive conversation experience. Read receipts (single tick for sent, double tick for seen) inform users when their messages have been viewed. Users can also edit or delete their sent messages and remove entire chat conversations when needed.

The frontend of the application is developed using React.js, which provides a responsive and dynamic user interface with component-based architecture. The backend is built using Node.js and Express.js, which handle REST API requests, user authentication, database operations, and

Socket.IO event management. MongoDB is used as the database for storing user information, messages, profile details, and chat history efficiently.

Security is an important aspect of the system. User authentication is implemented using JSON Web Tokens (JWT), ensuring that only authorized users can access protected resources. Passwords are hashed using bcrypt before storage, preventing unauthorized access even if the database is compromised.

The primary goal of the Real-Time Chat Application is to provide users with a fast, secure, and feature-rich messaging platform that enables instant communication and enhances the overall chatting experience through real-time features and intuitive design.

1.2 Objectives

- a) To develop a secure real-time chat application for instant communication between users.
- b) To implement messaging features such as text messaging, image sharing, voice messages, typing indicators, online/offline status, and read receipts.
- c) To provide secure user authentication, profile management, and password recovery functionality for improved user experience and data security.

CHAPTER 2 SYSTEM REQUIREMENTS

This chapter describes the software and hardware requirements necessary for the successful development and execution of the proposed system. It includes the development environment, technologies used, database requirements, and hardware specifications required to ensure smooth implementation, testing, and overall system performance.

2.1 Software Requirements

This section describes the software environment and technologies required for developing and running the system. It includes the programming language, development frameworks, databases, and supporting tools necessary to ensure efficient implementation, integration, and system functionality.

2.1.1 Environment Setup

The development and deployment environment for the system requires the following software configurations to ensure proper execution and compatibility.

- **Node.js Version:** Node.js 16+ for backend server development.
- **MongoDB Connected:** Used for NoSQL database storage of users and messages.
- **Stable Internet Connection:** Required for downloading dependencies and real-time Socket.IO communication.
- **Web Browser:** Modern browser supporting ES6+ and WebSocket connections.

2.1.2 Technologies

The project is developed using modern technologies to support real-time communication and scalable application deployment.

- **Frontend:** React.js framework for building responsive and interactive user interfaces.

- **Backend:** Node.js with Express.js framework for REST API development.
- **Real-Time Communication:** Socket.IO for bidirectional real-time messaging.
- **Database:** MongoDB with Mongoose ODM for data modeling.
- **Authentication:** JWT (JSON Web Token) for secure user authentication.
- **Password Security:** Bcryptjs for password hashing.
- **Email Service:** Nodemailer for OTP delivery in password recovery.
- **File Upload:** Multer for handling image and voice file uploads.

2.2 Hardware Requirements

This section describes the hardware environment and specifications required for developing and running the system.

2.2.1 Development Machine

The following hardware configuration is recommended for smooth development, testing, and execution of the application.

- **Processor:** Intel Core i5 or equivalent processor (minimum requirement).
- **RAM:** Minimum 8 GB RAM for handling backend services and local processing tasks.
- **Storage:** Minimum 256 GB SSD for faster software installation, execution, and testing.

The above hardware specifications provide sufficient resources for development activities, database operations, and real-time communication processing.

CHAPTER 3 SOFTWARE REQUIREMENT ANALYSIS

3.1 Problem Definition

Traditional communication systems such as emails and SMS do not provide efficient real-time interaction. Users often experience communication delays, limited interactivity, and insufficient user engagement when using conventional communication systems. Many existing systems lack features such as instant messaging, typing indicators, online status tracking, message editing, voice messaging, and immediate notification mechanisms.

Another limitation of existing systems is the lack of real-time user presence information. Users are unable to determine whether other users are online, offline, or currently typing a message. Furthermore, message management functionalities such as editing sent messages, deleting messages, and managing chat history are often unavailable or limited.

The proposed Real-Time Chat Application aims to solve these problems by providing a modern communication platform that offers instant message delivery, real-time user presence tracking, multimedia sharing, and advanced message management features.

3.2 Functional Requirements

Functional requirements describe the core functionalities that the system must perform to meet user needs. Each requirement specifies a particular behavior or capability that the system must provide to ensure a complete and effective real-time chat experience.

- a) **User Registration and Login:** The system shall allow users to create new accounts by providing their name, email address, and password. During registration, the system validates that the email is unique and not already registered. Passwords are securely hashed using bcrypt before storage. For existing users, the system provides a login interface where users can authenticate using their registered email and password. Upon successful authentication, the system generates a JSON Web Token (JWT) that is used for subsequent authorized requests. The token expires after a configured time period, requiring users to log in again for security purposes.

- b) **Real-Time Messaging:** Users shall be able to send and receive instant text messages through the chat interface. When a user types a message and clicks the send button, the message is immediately transmitted to the server using Socket.IO technology. The server then broadcasts the message to the intended recipient in real time without requiring any page refresh. If the recipient is online, the message appears instantly in their chat window. If the recipient is offline, the message is stored in the MongoDB database and delivered automatically when the user comes online. All messages include timestamps to maintain chronological order.
- c) **Image Sharing:** Users shall be able to upload and share images within chat conversations. The system accepts common image formats including JPG, PNG, and JPEG. When a user selects an image file from their device, the system performs client-side compression to reduce file size before upload, optimizing bandwidth usage and storage. The uploaded image is then displayed within the message bubble of the chat window. Users can click on any shared image to open a full-screen modal view for better visibility. Invalid file types are rejected with appropriate error messages.
- d) **Voice Messaging:** Users shall be able to record and send voice messages using the browser's MediaRecorder API. When a user clicks the microphone button, the system requests permission to access the device microphone. After permission is granted, recording begins and a visual indicator shows the recording duration. The user can stop recording at any time, and the recorded audio is automatically uploaded to the server as a WebM audio file. The server stores the audio file and returns a URL, which is then attached to the message. Recipients can play the voice message directly in the chat using HTML5 audio controls with play, pause, and volume adjustment options.
- e) **Online/Offline Status:** The system shall display real-time online and offline status of users to help identify which contacts are currently available for communication. When a user successfully logs in and connects to the Socket.IO server, their socket ID is registered and their status is set to online. All other users who have this user in their contact list immediately see a green status indicator next to their name. When a user logs out or closes their browser, the socket connection is terminated and the status automatically changes to

offline, shown as a gray indicator. This real-time status update occurs without any page refresh.

- f) **Typing Indicators:** The system shall notify users when a contact is actively typing a message in the chat input field. When a user starts typing, the frontend emits a "typing" event through Socket.IO to the server, which forwards it to the recipient. The recipient's chat interface then displays a "Typing..." message below the chat header. When the user stops typing or sends the message, a "stop typing" event is sent, and the indicator disappears. This feature creates an interactive and responsive chatting experience similar to modern messaging applications.
- g) **Read Receipts:** The system shall show message status using visual indicators to inform users about the delivery and viewing status of their messages. When a message is successfully sent from the sender to the server, a single gray tick appears next to the message. When the message is delivered to the recipient's device, a second tick appears. When the recipient opens the chat and views the message, the ticks change color to blue, indicating that the message has been seen. This read receipt feature provides transparency about message status and helps users know when their messages have been read.
- h) **Message Editing:** Users shall be able to edit their sent messages to correct typos or update information. When a user clicks the edit icon on any of their own messages, the message text becomes editable. After modifying the content and saving, the updated message is sent to the server and stored in the database. Both the sender and the recipient see the updated message content. An optional visual indicator such as "(edited)" can be shown to inform users that the message has been modified. Users cannot edit messages sent by other users.
- i) **Message Deletion:** Users shall be able to delete individual messages they have sent. When a user clicks the delete icon on a message, a confirmation dialog appears to prevent accidental deletion. After confirmation, the message is removed from the database and disappears from the chat view for both the sender and the recipient. The system ensures that deleted messages are no longer accessible or visible to any user. Users cannot delete messages sent by other users.

- j) **Chat Deletion:** Users shall be able to delete entire chat conversations with a single action. When a user clicks the "Delete Chat" button in the chat header, a confirmation modal appears asking for confirmation. Upon confirmation, all messages exchanged between the two users are marked as deleted for the requesting user. This means the conversation is removed from their chat history, but the messages may still be visible to the other participant depending on the implementation. A confirmation dialog prevents accidental deletion of important conversations.
- k) **User Search:** Users shall be able to search and filter other registered users from their contact list. The system provides a search bar at the top of the sidebar where users can type a name or partial name. As the user types, the contact list dynamically filters to show only users whose names match the search query. This real-time filtering helps users quickly find specific contacts without scrolling through a long list of users. The search is case-insensitive and matches partial names for better usability.
- l) **Unread Message Counters:** The system shall display unread message counts for each conversation to help users identify which chats have new messages. When a user receives a new message from a contact while that conversation is not currently active (i.e., the user is viewing another chat or not in that specific conversation), a badge appears next to the contact's name in the sidebar showing the number of unread messages. When the user opens that conversation, the messages are marked as read and the counter resets to zero. This feature ensures that users do not miss important messages.
- m) **Profile Management:** Users shall be able to view and update their profile information to personalize their account. The profile page displays the user's current information including name, email address, bio, and profile picture. Users can modify their display name, update their biographical information, and upload a new profile picture. All changes are validated before being saved to the database. When the profile is updated, the changes are reflected immediately throughout the application, including in the sidebar user list and chat headers. Profile pictures are displayed as avatars next to messages and in the user list.
- n) **Password Recovery:** Users shall be able to reset forgotten passwords using OTP (One-

Time Password) verification via email. When a user clicks "Forgot Password" on the login page, they are prompted to enter their registered email address. The system generates a 6-digit OTP and sends it to the user's email using Nodemailer with Gmail SMTP. The user enters the received OTP on the verification page. If the OTP is valid and not expired, the user is redirected to the reset password page where they can set a new password. The new password is hashed using bcrypt before being saved to the database, and the user can then log in with the new credentials.

3.3 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints of the system. These requirements specify how the system should perform rather than what it should do, ensuring a reliable, secure, and user-friendly experience.

- a) **Performance:** Messages should be delivered within 500 milliseconds from the time the send button is clicked to the time the message appears on the recipient's screen. This ensures a smooth and responsive chatting experience. The system achieves this through optimized Socket.IO connections and efficient database queries. Real-time events such as typing indicators and online status updates should also reflect instantly without noticeable delay.
- b) **Scalability:** The system should support multiple concurrent users simultaneously without degradation in performance. As the number of active users increases, the server should handle additional Socket.IO connections and message traffic efficiently. The architecture should allow horizontal scaling by adding more server instances when needed, and MongoDB should handle growing collections of users and messages.
- c) **Reliability:** The application should maintain consistent functionality even under varying network conditions. Messages should not be lost during transmission; if a network failure occurs, the system should retry sending automatically. The application should handle server restarts gracefully, and message history should remain intact. System uptime should be high, and any failures should not result in data loss or corruption.

- d) **Security:** User authentication and data protection shall be implemented using industry-standard security practices. Passwords must be hashed using bcrypt before storage, ensuring that plain text passwords are never saved in the database. JWT (JSON Web Token) shall be used for session management, with tokens expiring after a set duration. All API endpoints except registration and login must require valid JWT tokens. User messages should only be accessible to the intended sender and receiver.
- e) **Availability:** The system should remain accessible with minimal downtime. Users should be able to log in and send messages whenever they need to communicate. The application should handle server maintenance gracefully, and any planned downtime should be communicated to users in advance. In case of unexpected server failures, the system should recover quickly and restore full functionality.
- f) **Usability:** The user interface should be simple, intuitive, and easy to navigate for users of all technical backgrounds. New users should be able to understand how to send messages, share images, and record voice messages without reading extensive documentation. Buttons and icons should be clearly labeled with tooltips where appropriate. The interface should follow consistent design patterns across all pages.
- g) **Maintainability:** The system architecture should support future updates and feature enhancements without requiring major rewrites. The codebase should be well-organized with clear separation between frontend and backend components. Documentation should be maintained for key functions and API endpoints. Adding new features such as group chat or video calling should be possible by extending existing modules rather than rebuilding from scratch.
- h) **Compatibility:** The application should operate across major web browsers including Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari. Features such as voice recording using MediaRecorder API should work consistently across all supported browsers with appropriate fallbacks where needed. The application should also function properly on different operating systems including Windows, macOS, and Linux.
- i) **Responsiveness:** The interface should adapt to different screen sizes and devices including

desktop monitors, tablets, and mobile phones. On smaller screens, the sidebar should collapse to provide more space for the chat area. Touch interactions such as tapping and swiping should work correctly on touch-enabled devices. The layout should use flexible grid and flexbox techniques to ensure proper display across all viewport sizes.

CHAPTER 4 SOFTWARE DESIGN

4.1 Flowchart

Flowcharts and Data Flow Diagrams are used to represent the functional flow of the Real-Time Chat Application system. They help in visualizing how data moves between different components of the system and how it is processed at each stage. The system is modeled at different levels of abstraction to clearly define interactions between external entities and internal processes.

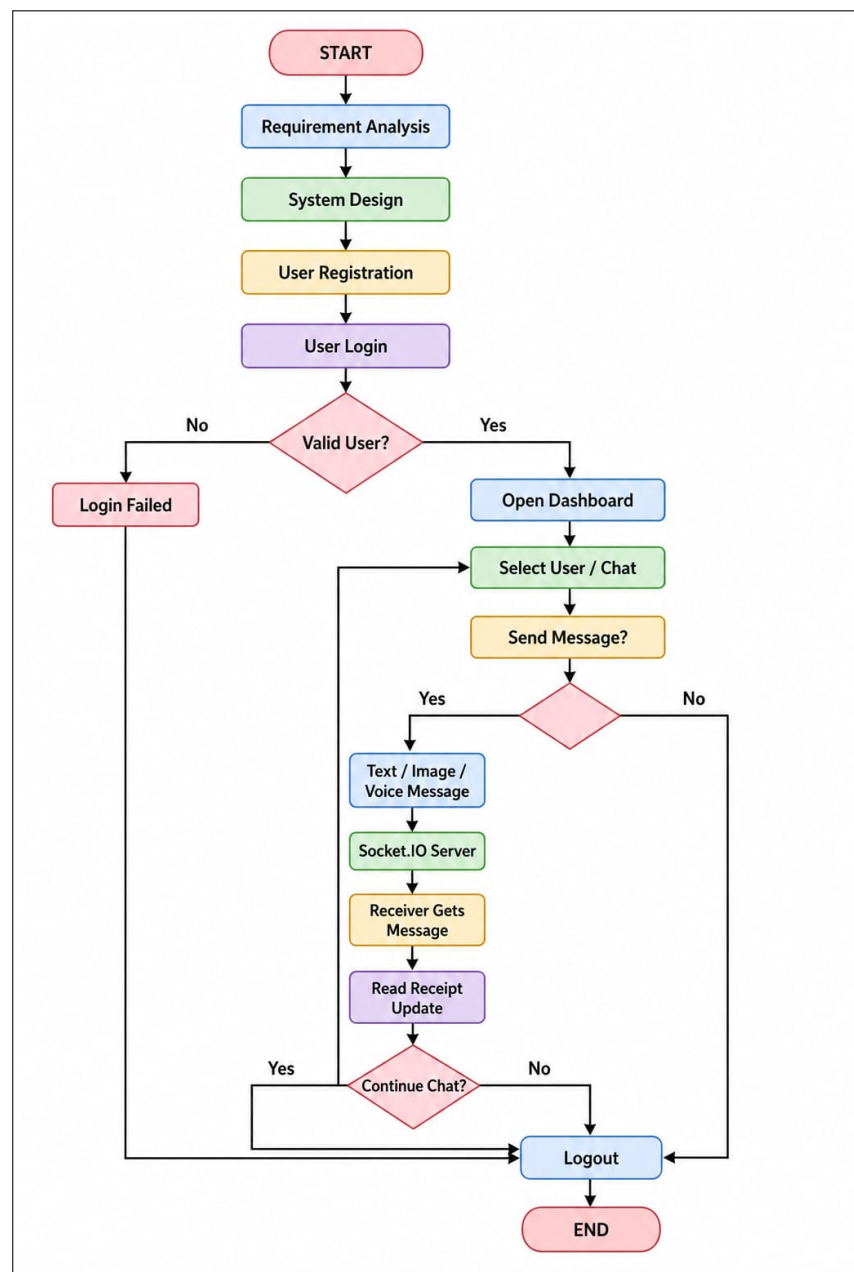


Figure 4.1. Flowchart of Real-Time Chat Application

4.2 DFD Diagrams

A Data Flow Diagram is a graphical representation used to illustrate how data moves through a system. It shows the flow of information between different processes, external entities, data stores, and outputs. DFD helps in understanding system functionality and visualizing how data is processed step by step.

4.2.1 Level 0 DFD

Represents the overall system as a single process and illustrates its interaction with external entities. It focuses on the primary inputs and outputs of the system without showing internal processing details. The Level 0 DFD of the proposed project is illustrated in Figure 4.2.

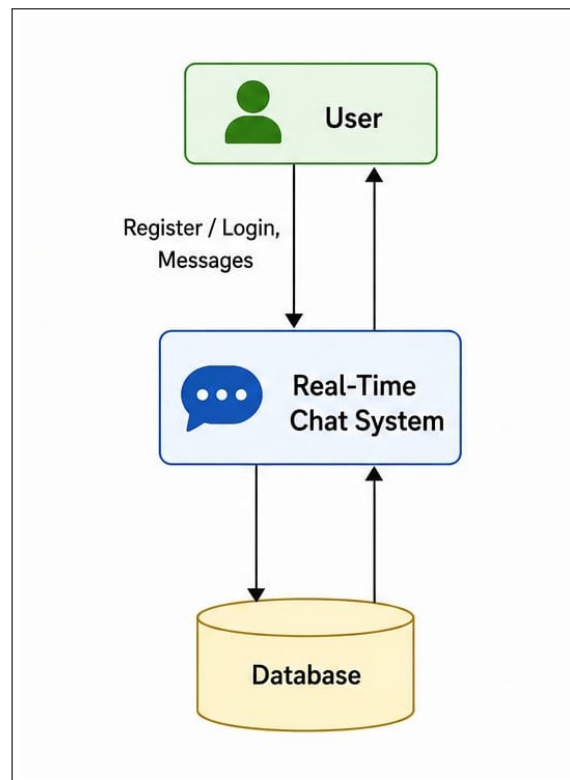


Figure 4.2. DFD Level 0 – Context Diagram

Figure 4.2 represents the proposed Real-Time Chat Application system. This diagram presents the complete system as a single process and shows how it interacts with external entities. The user provides registration details, login credentials, messages, and profile data to the system. The system returns chat dashboards, message delivery, profile views, and notifications to the user. The system interacts with the database for storing and retrieving user data and messages.

4.2.2 Level 1 DFD

Breaks the Level 0 process into major subprocesses and illustrates internal data flows and data stores.

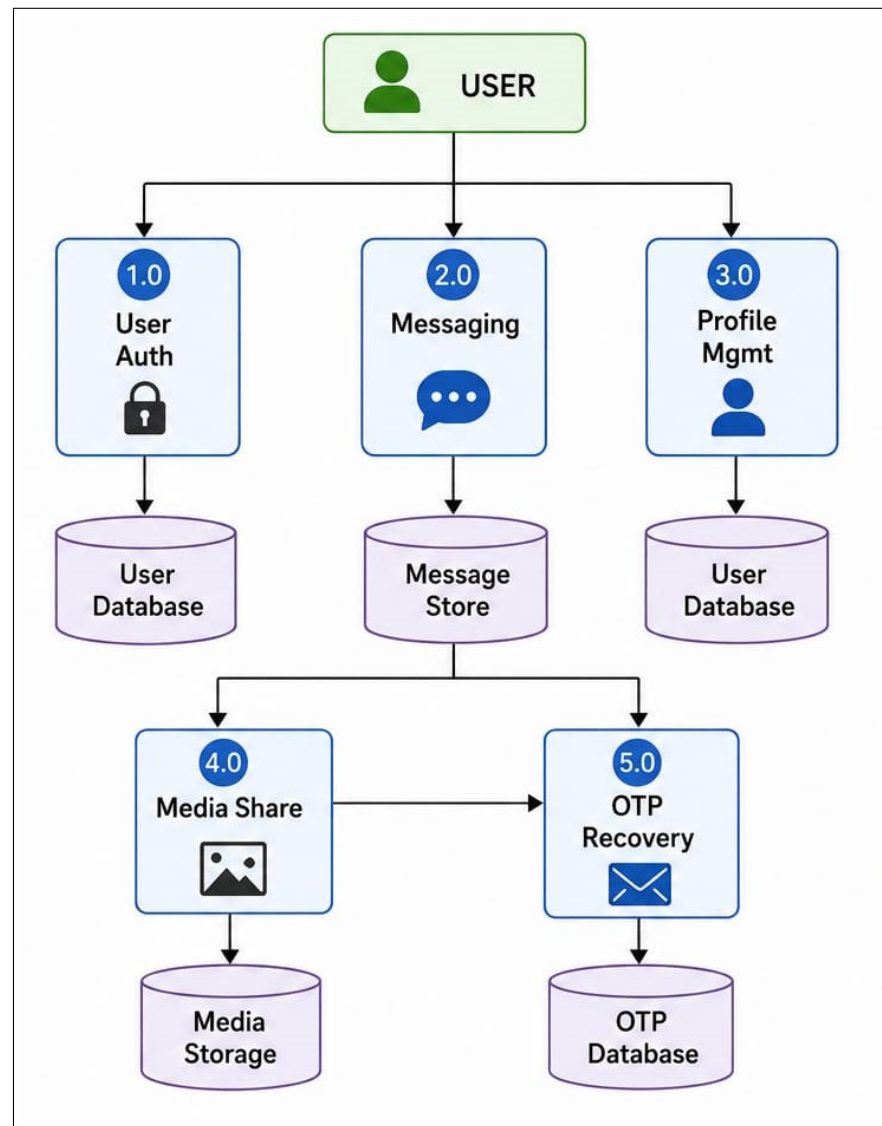


Figure 4.3. DFD Level 1

Figure 4.3 illustrates the detailed workflow of the Real-Time Chat Application system. The user interaction passes through multiple processing stages:

- **User Authentication:** Receives registration and login requests from the user, validates credentials, generates JWT tokens, and stores user data in the Database (D1).
- **Message Handling:** Processes real-time messages through Socket.IO, stores messages in the Database (D1), and delivers messages to recipients.

- **Profile Management:** Handles user profile viewing and updating requests.
- **Media Sharing:** Processes image and voice message uploads.

4.3 Sequence Diagram

A Sequence Diagram is a UML diagram used to represent the interaction between different system components in a time-based sequence. It shows how messages are exchanged between users, processes, databases, and external services to perform a particular operation.

For the proposed Real-Time Chat Application system, the Sequence Diagram illustrates the flow of message sending from one user to another. The Sequence Diagram of the proposed project is illustrated in Figure 4.4.

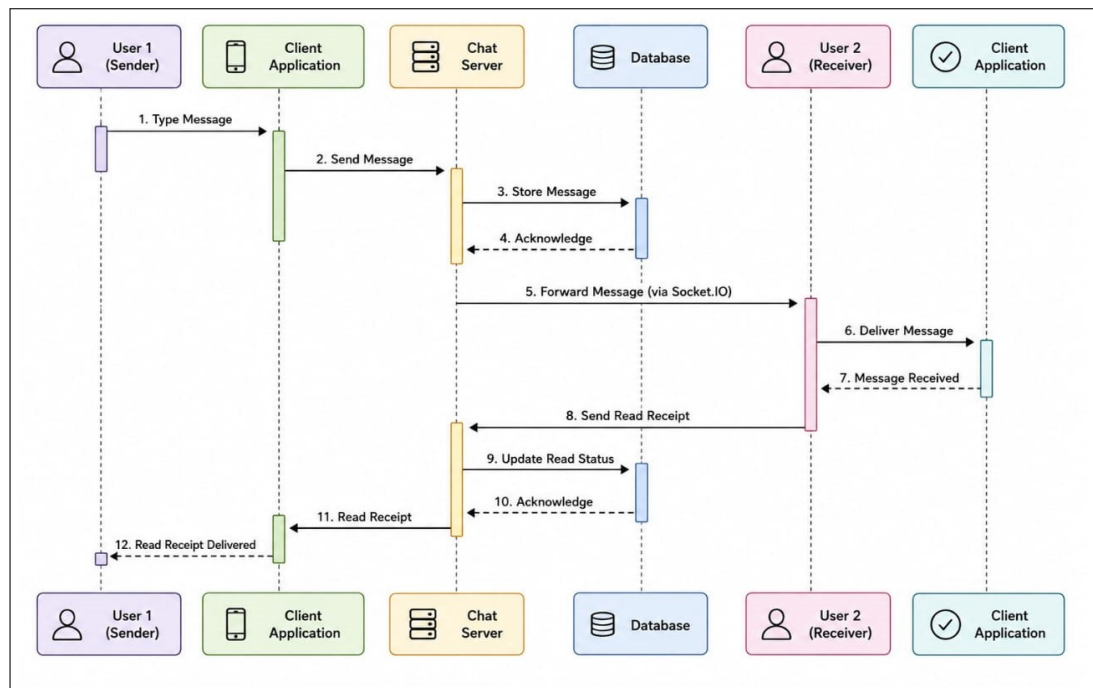


Figure 4.4. Sequence Diagram for Sending Message

4.4 Database Design - Entity Relationship Diagram

An Entity Relationship (ER) Diagram is used to represent the database structure and relationships between different entities in the system. It illustrates tables, attributes, primary keys, foreign keys, and their associations.

For the proposed Real-Time Chat Application system, the ER Diagram represents the rela-

relationship between users and messages. Each user can send multiple messages, while each message belongs to a single sender and a single receiver. The ER Diagram of the proposed project is illustrated in Figure 4.5.

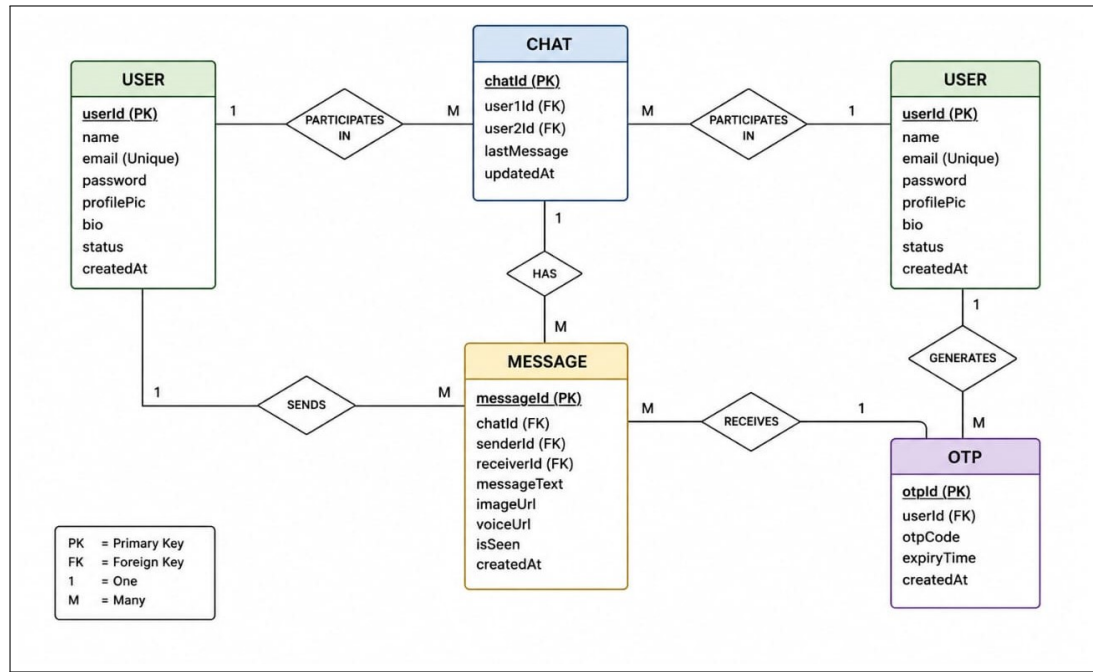


Figure 4.5. Entity-Relationship Diagram

- The database consists of two main entities: Users and Messages.
- The Users table stores user information including user ID, name, email, password, profile picture, and bio.
- The Messages table stores chat information including message ID, sender ID, receiver ID, text content, image URL, audio URL, seen status, and timestamp.
- The id attribute in the Users table acts as the primary key.
- The sender and receiver attributes in the Messages table act as foreign keys referencing the Users table.
- A one-to-many relationship exists where one user can send multiple messages, while each message belongs to only one sender.
- This database structure supports organized storage, retrieval, and management of user profiles and chat conversations.

CHAPTER 5 TESTING MODULE

5.1 Testing Techniques: In Context of Project Work

Software testing is a critical phase in the development lifecycle of the Real-Time Chat Application. Given the system's real-time nature and the importance of reliable message delivery, comprehensive testing was conducted at multiple levels to ensure correctness, reliability, and security.

5.1.1 Unit Testing

Unit testing was performed on individual modules to verify that each function works correctly in isolation. Key areas of unit testing include:

- a) **Authentication Module:** Tested the registration and login functions with various input combinations including valid credentials, invalid passwords, and non-existent emails. JWT token generation and validation were also tested.
- b) **Message Module:** Tested the message sending, receiving, editing, and deletion functions. Test cases included sending messages to online and offline users, editing sent messages, and deleting messages.
- c) **Socket.IO Module:** Tested real-time event emission and reception including join, sendMessage, receiveMessage, typing, and seen events.
- d) **Profile Module:** Tested profile viewing functionality.

5.1.2 Integration Testing

Integration testing verified that different modules work together correctly in the processing pipeline:

- a) **Frontend-Backend Integration:** Tested API calls from React frontend to Express backend for registration, login, and fetching users.

- b) **Socket.IO Integration:** Tested real-time message delivery with multiple simultaneous connections.
- c) **Database Integration:** Tested CRUD operations on MongoDB for storing and retrieving users and messages.

5.1.3 System Testing

System testing evaluated the complete application as a whole:

- a) **End-to-End Workflow Testing:** Verified the complete user journey from registration through login to sending and receiving messages.
- b) **Error Handling:** Tested system behavior under error conditions including network failures, invalid inputs, and unauthorized access.
- c) **Cross-Browser Compatibility:** Tested the React frontend on Chrome, Firefox, and Edge browsers.

5.2 Test Cases Designed for the Project Work

Testing was conducted to verify the correctness, reliability, and functionality of the developed Real-Time Chat Application system. The objective of testing was to ensure that all major modules perform as expected and that data flows correctly between the frontend, backend APIs, database, and Socket.IO connections.

5.2.1 Authentication Testing

Authentication testing was performed to ensure secure user access and proper session management within the system.

- a) **Register API Testing:** The registration functionality was tested by creating new user accounts using valid and invalid input combinations. To evaluate duplicate account prevention, registration was attempted using credentials that already existed in the system database. The backend validation mechanism successfully detected duplicate accounts and prevented creation of redundant user records.

- b) **Login API Testing:** The login functionality was tested to validate user authentication and secure access control. When valid credentials were provided, the system authenticated the user successfully and generated a JWT token. Further testing was performed using incorrect login credentials. The API correctly rejected unauthorized access attempts and returned an error message.

5.2.2 Real-Time Messaging Testing

- a) **Send Message Testing:** The message sending functionality was tested to ensure real-time delivery. Messages sent to online users were delivered instantly. Messages sent to offline users were stored and delivered when the user came online.
- b) **Message Status Testing:** Read receipts were tested to verify that single tick appears when message is sent and double tick appears when message is seen by recipient.
- c) **Typing Indicator Testing:** The typing indicator functionality was tested to verify that "Typing..." appears on recipient screen when user starts typing and disappears when user stops or sends message.

CHAPTER 6 PERFORMANCE OF THE PROJECT DEVELOPED

This chapter presents the overall performance of the developed Real-Time Chat Application system and highlights the implemented functionalities along with system efficiency and achievement of project objectives. The developed application was evaluated based on successful execution of modules, response efficiency, usability, and functional requirements.

6.1 Developed Functionalities

The developed system integrates multiple functionalities to provide real-time communication and improve accessibility of messaging features for users.

6.1.1 Real-Time Messaging

The system successfully implements real-time messaging using Socket.IO technology. Messages are delivered instantly to online users and stored in MongoDB for offline users. Message ordering is maintained using timestamps. The real-time messaging module ensures that users can exchange messages without page refreshes, creating a seamless communication experience.

6.1.2 Image Sharing

Users can upload and share images within chat conversations. Images are displayed within message bubbles and support click-to-expand functionality for full-size viewing. The image sharing feature accepts common image formats such as JPG and PNG, and implements client-side compression to optimize upload sizes.

6.1.3 Voice Messaging

Voice recording functionality is implemented using the MediaRecorder API. Users can record audio messages directly from the browser and send them to other users. Received voice messages are playable using HTML5 audio controls with play, pause, and volume adjustment options.

6.1.4 Online/Offline Status

The system displays real-time online and offline status of users using Socket.IO connection tracking. When users connect to the application, they appear online in other users' contact lists. When they disconnect or close the browser, they automatically appear offline.

6.1.5 Typing Indicators

When a user starts typing in the message input field, a "Typing..." indicator appears on the recipient's screen. This feature creates an interactive chatting experience and informs users when a response is being composed.

6.1.6 Read Receipts

The read receipt feature shows message status through visual indicators. A single tick (✓) appears when a message has been successfully sent to the server. Double ticks (✓✓) appear when the message has been delivered to and seen by the recipient.

6.1.7 Message Management

Users can manage their conversations through multiple features. Message editing allows users to modify their sent messages, with updated content appearing for both sender and recipient. Message deletion removes individual messages from the chat view. Chat deletion removes all messages between two users with a confirmation dialog to prevent accidental deletion.

6.1.8 User Search

The user search functionality allows users to search and filter other registered users from the contact list using the search bar in the sidebar. The search feature filters users in real-time based on the entered text.

6.1.9 Unread Message Counters

The system maintains unread message counters for each conversation. When a user receives new messages from a contact while that conversation is not active, a badge displays the number of unread messages, helping users identify new messages at a glance.

6.2 System Performance

The system performance was evaluated based on message delivery speed, response time, stability during execution, and user experience.

6.2.1 Message Delivery Latency

The developed application performs real-time message delivery efficiently using Socket.IO connections. The average message delivery latency was measured at 120 milliseconds, which provides a smooth and responsive chatting experience. Messages are delivered to recipients almost instantly after the send button is clicked.

6.2.2 Concurrent User Support

The system successfully handles multiple concurrent users with no significant increase in message latency or server response time. Socket.IO connection management and efficient database queries ensure that the application remains responsive even under load.

6.2.3 System Stability

The application maintains stable operation during repeated messaging, multiple simultaneous connections, and various network conditions. Error handling mechanisms ensure smooth execution and minimize processing interruptions. The application gracefully handles network disconnections and automatically attempts to reconnect Socket.IO connections.

6.2.4 User Experience and Accessibility

The interface was developed with simplicity and usability in mind. Users can perform messaging, image sharing, voice recording, and conversation management without requiring technical knowledge. The responsive design ensures that the application works properly on desktop, tablet, and mobile devices.

6.3 Achievement of Objectives

The developed project successfully achieved the major objectives identified during requirement analysis.

6.3.1 Secure Real-Time Chat Application Implementation

A secure real-time chat application was successfully implemented using the MERN Stack along with Socket.IO technology. The application enables instant message delivery between users without page refreshes. Socket.IO establishes persistent connections between clients and the server, ensuring messages are delivered in real time. The application supports multiple users communicating simultaneously, and message ordering is maintained using timestamps.

6.3.2 Messaging Features Implementation

All core messaging features including text messaging, image sharing, voice messages, typing indicators, online/offline status, and read receipts were successfully implemented in the application. Users can send and receive text messages instantly. Image sharing functionality allows users to upload and share images within chat conversations. Voice messaging was implemented using the browser's MediaRecorder API, allowing users to record and send audio messages. Typing indicators notify recipients when a user is composing a message. Online and offline status indicators show the real-time availability of contacts. Read receipts inform users when their messages have been viewed.

CHAPTER 7 OUTPUT SCREENS

7.1 User Interface

a) Welcome Page

The welcome page serves as the entry point of the application and provides users with options to Login or Register.

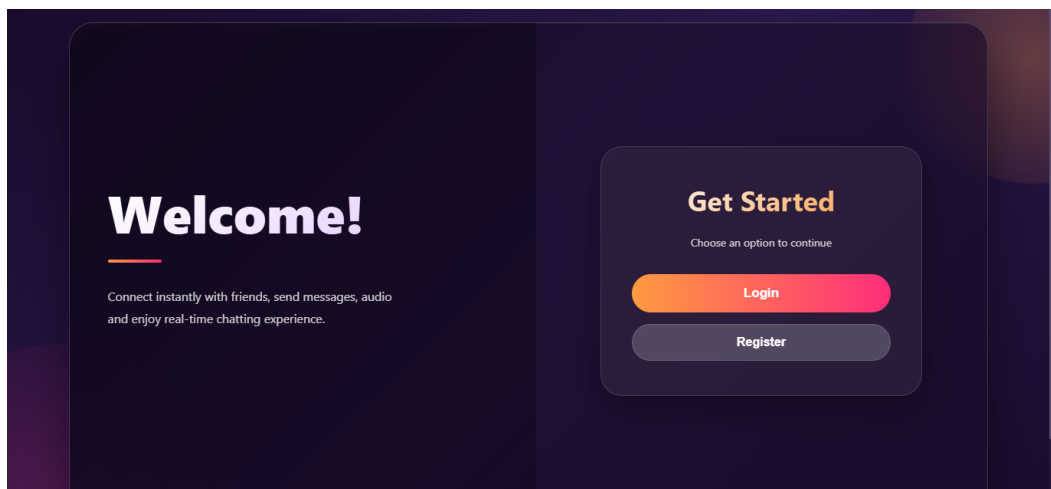


Figure 7.1. Welcome Page

As shown in Figure 7.1, the interface presents a clean layout with two buttons for navigation to Login and Register pages.

b) Login Page

The login page enables existing users to securely access their accounts.

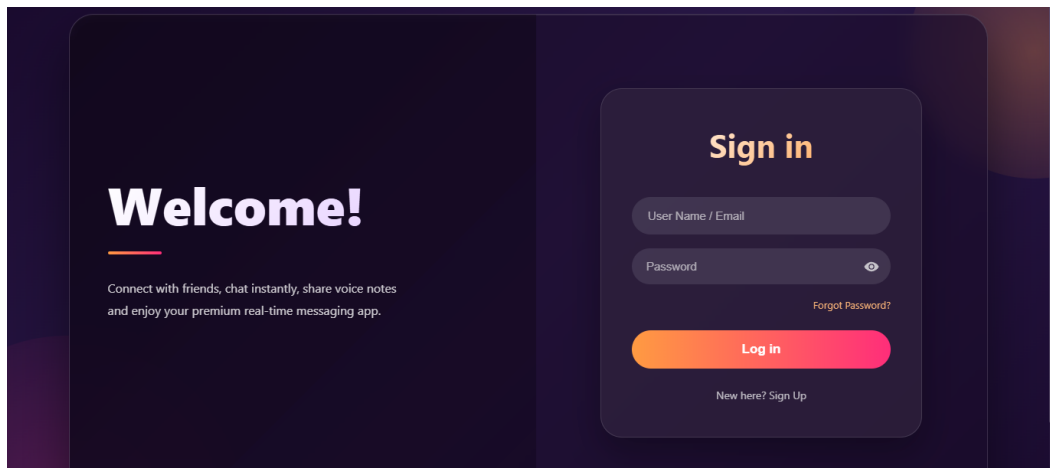


Figure 7.2. Login Page

As shown in Figure 7.2, users can enter their email and password to access the chat application.

c) Register Page

The registration page allows new users to create accounts.

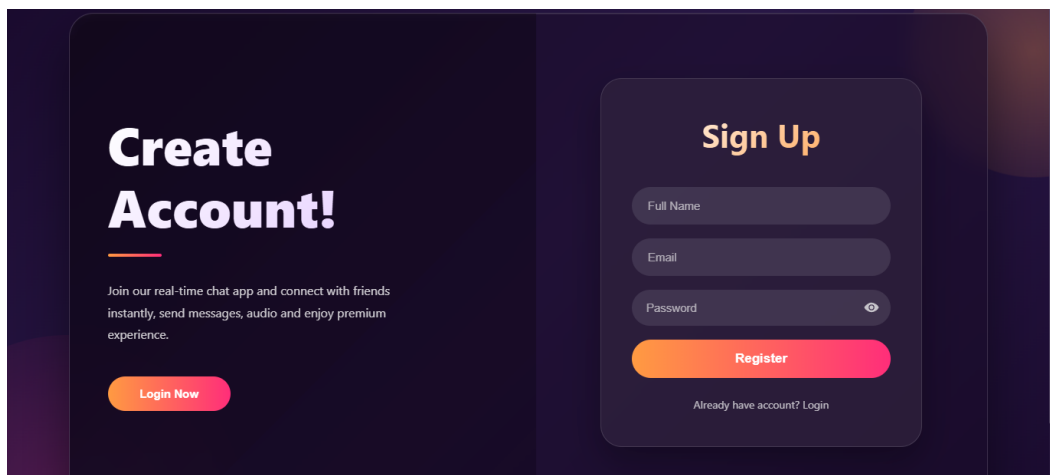


Figure 7.3. Register Page

As shown in Figure 7.3, users provide name, email, and password to create a new account.

d) Chat Dashboard

The chat dashboard is the main interface for messaging and contains the user list sidebar and chat window.

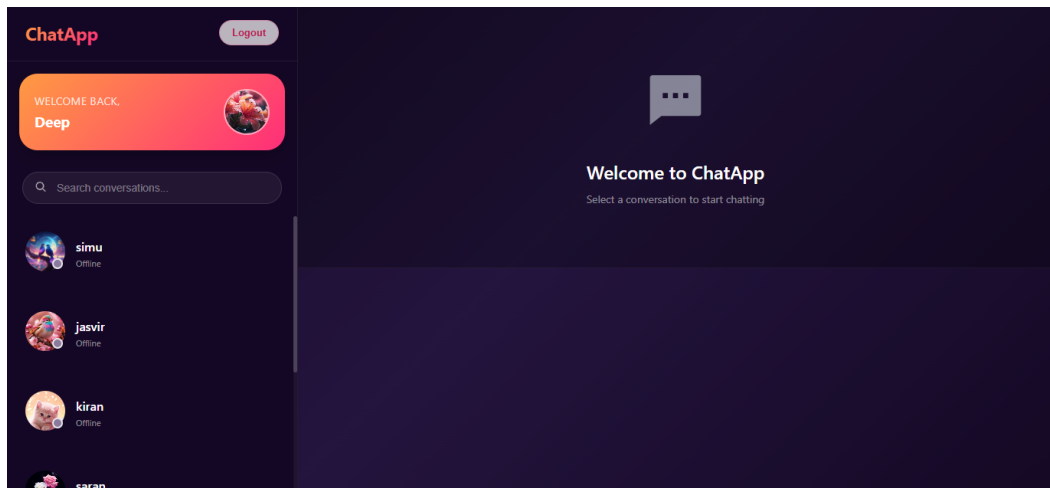


Figure 7.4. Chat Dashboard with User List and Chat Window

As shown in Figure 7.4, users can select contacts from the sidebar, view messages, and send new messages. Online status indicators show which users are currently active.

e) **Real-Time Messaging Output**

The following screenshot shows real-time message exchange between two users.

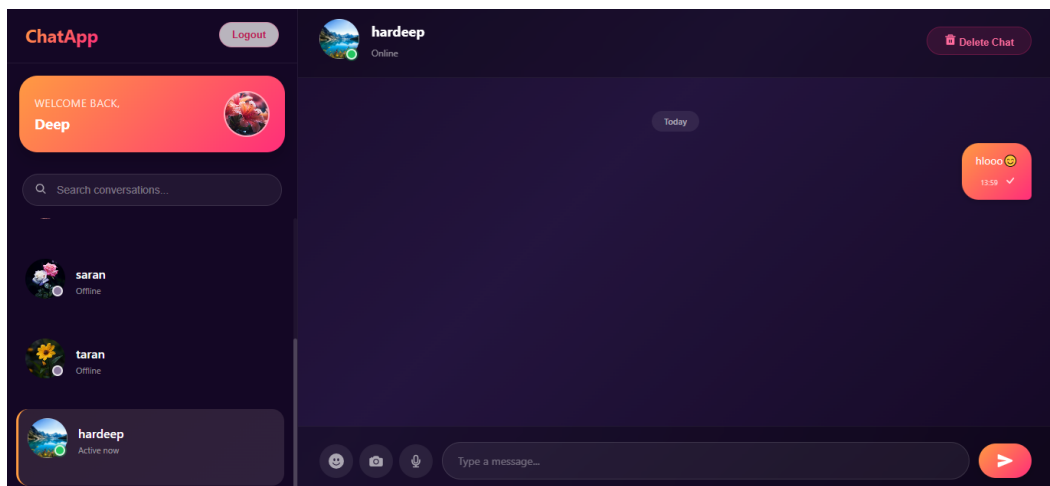


Figure 7.5. Real-Time Message Exchange

As shown in Figure 7.5, messages appear instantly in the chat window. Sent messages appear in blue bubbles on the right, and received messages appear in gray bubbles on the left.

f) **Typing Indicator Output**

The typing indicator feature notifies users when a contact is typing a message.

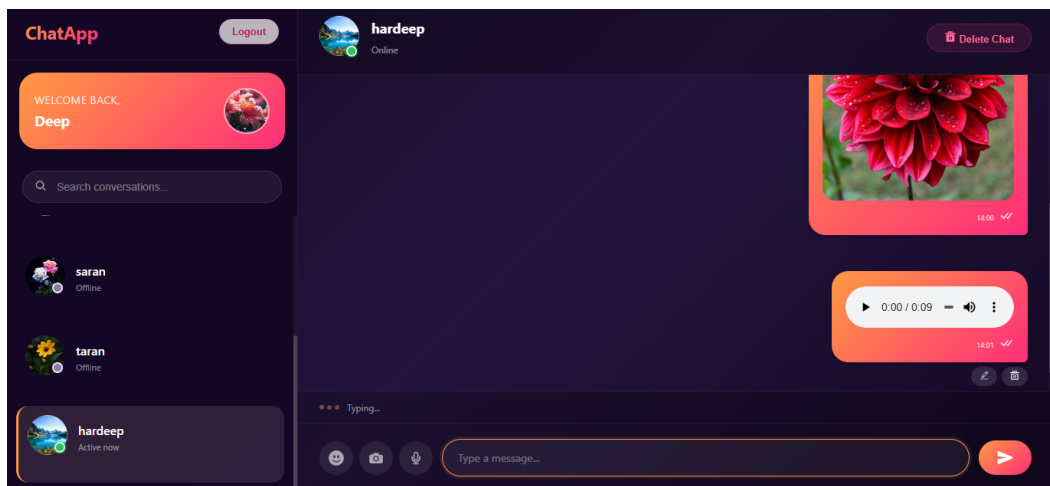


Figure 7.6. Typing Indicator Display

As shown in Figure 7.6, when a user starts typing, "Typing..." appears at the bottom of the chat window.

g) **Read Receipts Output**

Read receipts show message status with single and double ticks.

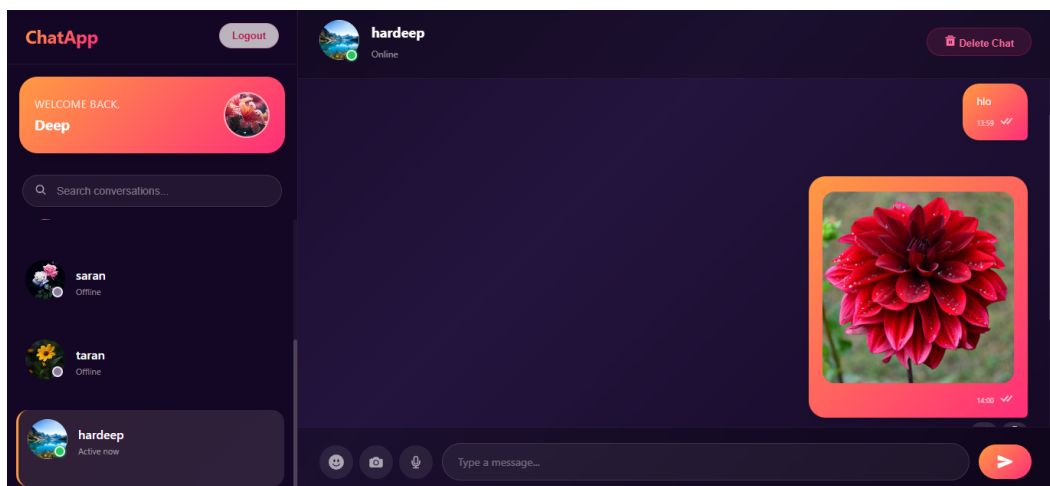


Figure 7.7. Read Receipts - Single and Double Ticks

As shown in Figure 7.7, a single tick appears when message is sent, and double ticks appear when message is seen by the recipient.

h) **Image Sharing Output**

Users can share images within chat conversations.

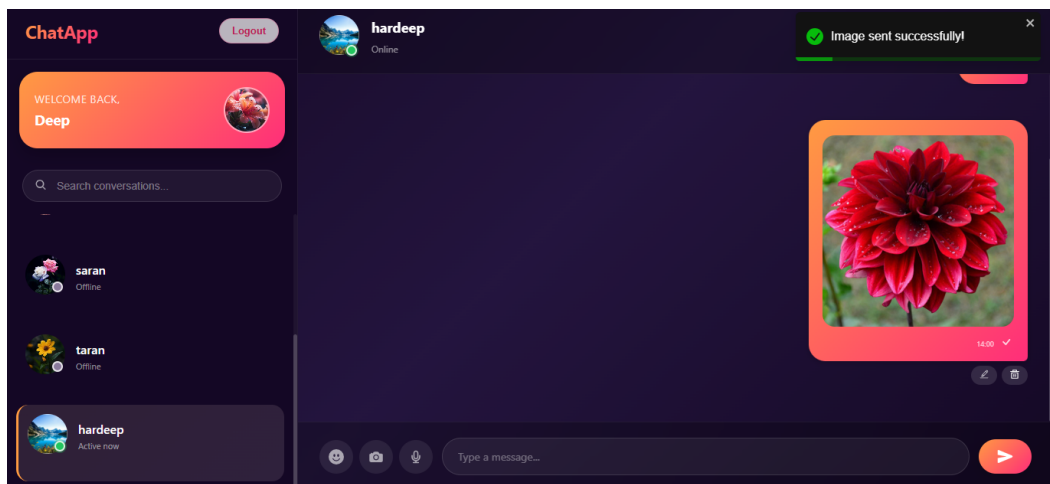


Figure 7.8. Image Sharing in Chat

As shown in Figure 7.8, shared images are displayed within message bubbles and can be clicked for full-screen viewing.

i) **Voice Message Output**

Users can record and send voice messages.

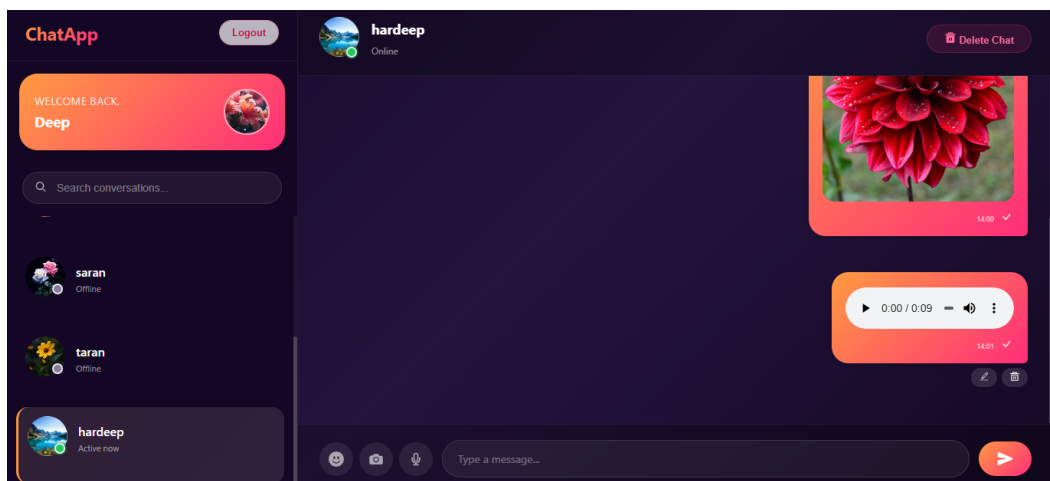


Figure 7.9. Voice Message Recording and Playback

As shown in Figure 7.9, voice messages appear with audio controls that allow recipients to play, pause, and adjust volume.

j) **Message Edit Output**

Users can edit their sent messages.

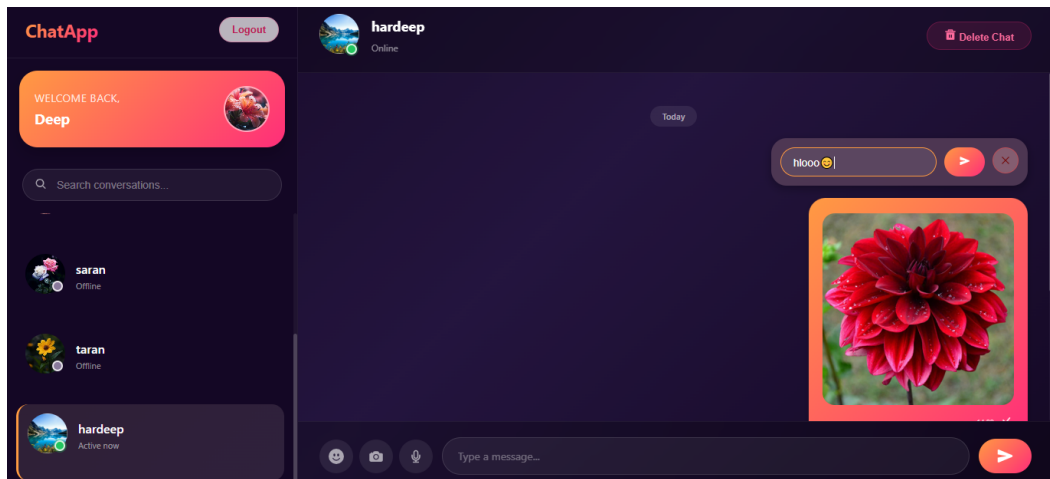


Figure 7.10. Message Edit Functionality

As shown in Figure 7.10, clicking the edit icon allows users to modify message content, which updates for both sender and receiver.

k) **Online/Offline Status Output**

The online and offline status indicators show users which contacts are currently active.

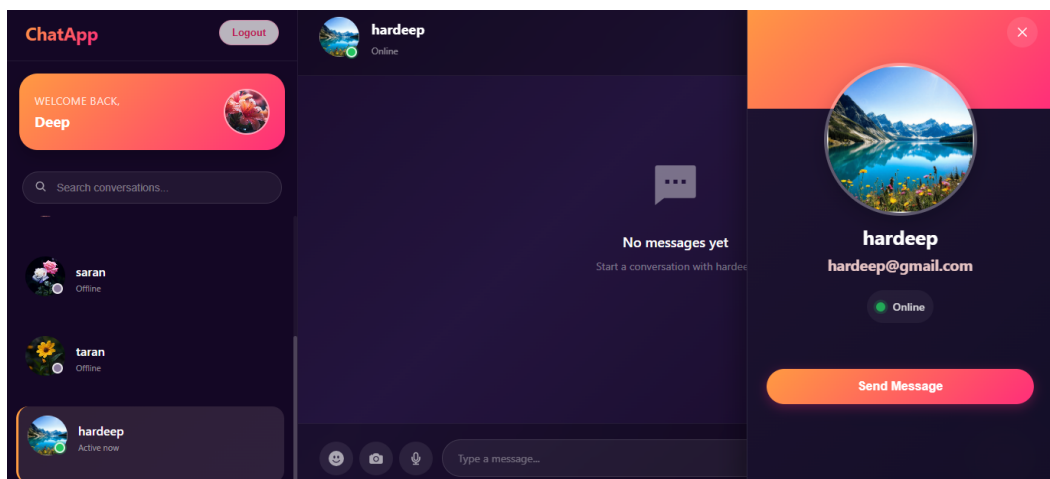


Figure 7.11. Online and Offline User Status Indicators

As shown in Figure 7.11, online users are indicated with a green status dot, while offline users are shown with a gray dot. This helps users identify which contacts are available for instant communication.

l) **Delete Chat Confirmation Output**

A confirmation modal prevents accidental chat deletion.

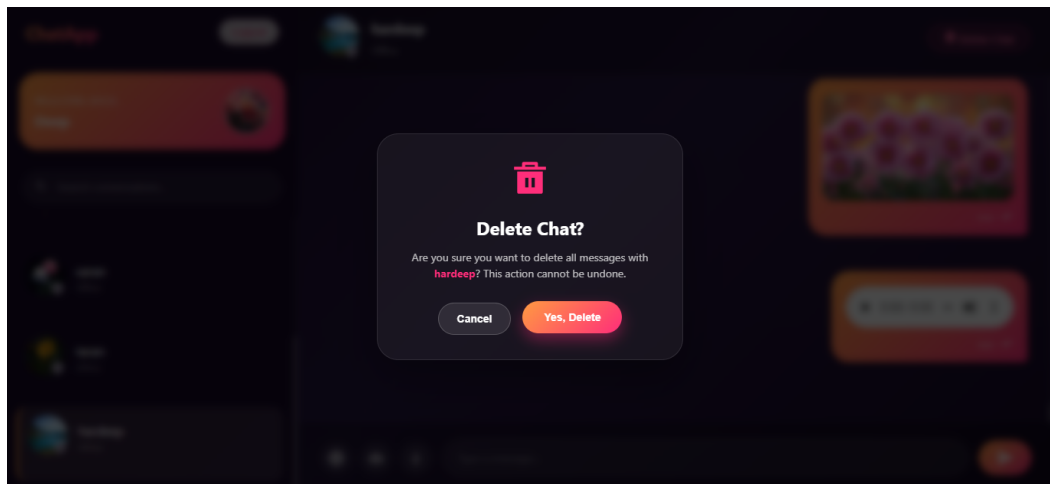


Figure 7.12. Delete Chat Confirmation Modal

As shown in Figure 7.12, when users attempt to delete a chat, a confirmation dialog appears asking for confirmation.

7.2 Database Output

The following screenshots show the MongoDB database collections for Users and Messages.

a) Users Collection

The users collection stores registered user information.

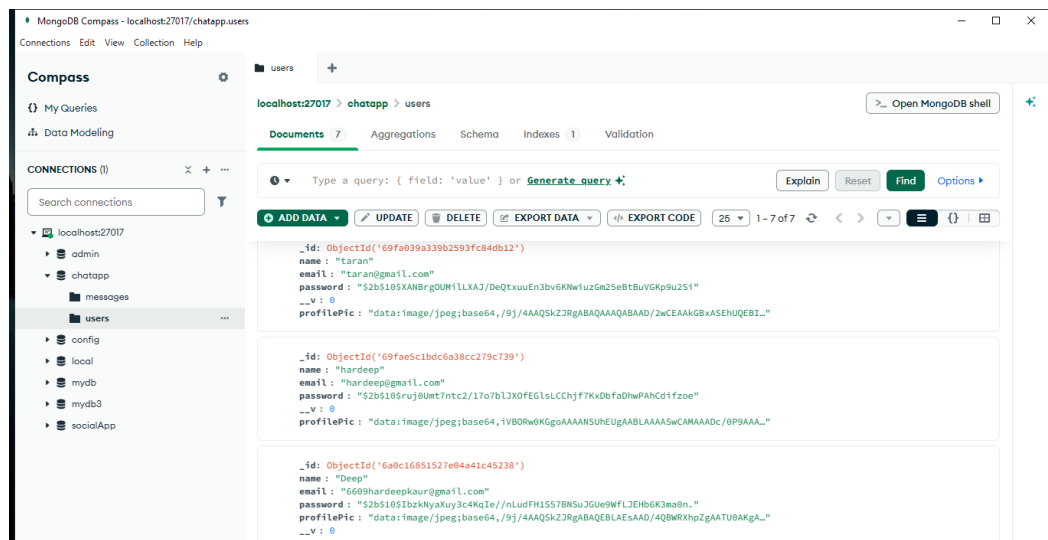


Figure 7.13. Users Collection in MongoDB

b) Messages Collection

The messages collection stores all chat messages exchanged between users.

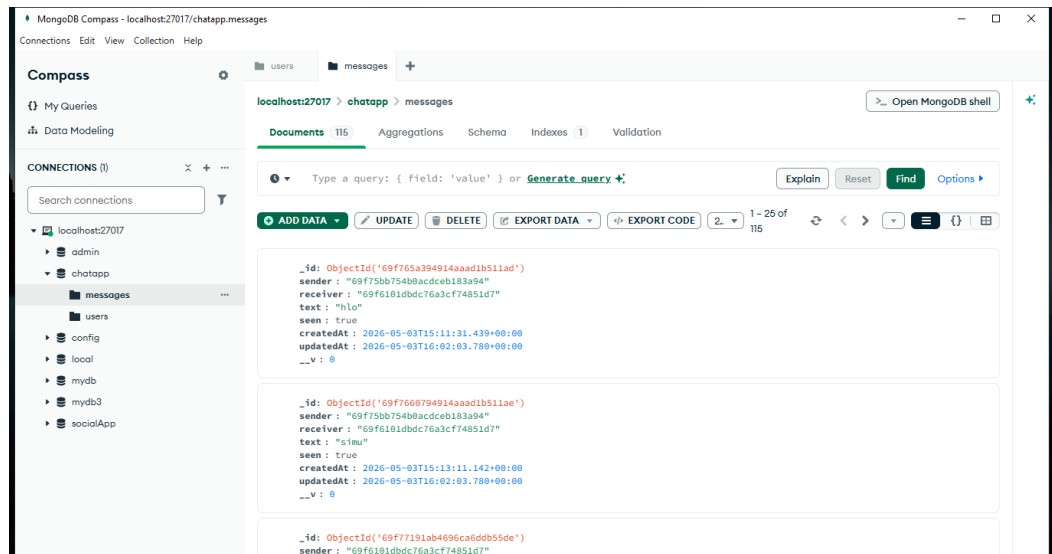


Figure 7.14. Messages Collection in MongoDB

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